

Measures of Dispersion

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1. Overview

Key point: A measure of dispersion says how spread out a column is around its centre: range, variance, mean absolute deviation, standard deviation and coefficient of variation each do this differently.

Variance: the average squared distance from the mean

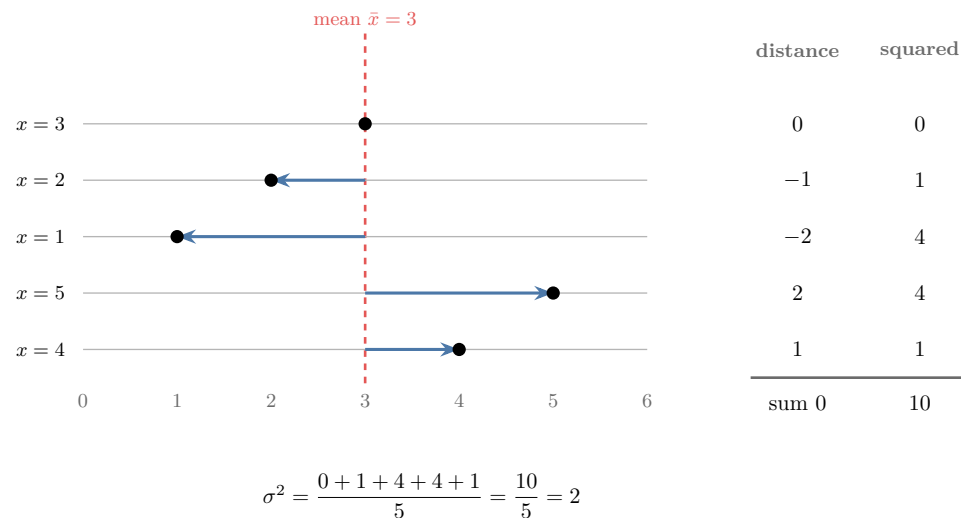


Figure 1: Variance of 3, 2, 1, 5, 4: each value's distance from the mean, squared and averaged

Figure 1 shows the idea behind the most used measure, the variance. This Note goes through five measures of spread in turn, works each with small numbers, and shows why the sample variance divides by $n - 1$.

2. Why the centre is not enough

Key point: Two columns can share a mean and still be spread very differently.

Take two columns of three values each: $-5, 0, 5$ and $-10, 0, 10$. Both have mean 0, yet the second is clearly more spread out, as the [PCA intuition Note](#) (section 5.1) shows. A measure of centre alone cannot tell them apart.

A **measure of dispersion** is a statistical measure that describes the spread, or variability, of a dataset: how the data is distributed around its centre.

3. Range

Key point: The range is the largest value minus the smallest; it is simple but ruined by a single outlier.

The **range** is the distance between the two extremes of a column.

1. **In words:** subtract the smallest value from the largest.
2. **Formula:**

$$\text{range} = \max(x) - \min(x)$$

3. **Example:** for $-5, 0, 5$ the range is $5 - (-5) = 10$; for $-10, 0, 10$ it is $10 - (-10) = 20$. The range tells the two columns apart where the mean could not.

The range uses only two values, the two extremes, so one outlier can blow it up. If a column's values all lie between 0 and 50 except one at 250, the range jumps from about 50 to 250 because of that one point.

Salaries in India show the problem. Some of the richest people in the world live in India, and so do many people who earn very little. The range of Indian salaries is therefore huge, yet it says nothing about how most people earn. So the range is rarely used on its own.

4. Variance

Key point: Variance is the average squared distance from the mean; the sample version divides by $n - 1$ instead of n .

The variance is defined in the [PCA intuition Note](#) (section 5) as the average squared distance of the values from their mean. Here we add the two versions of the formula, work them through, and see why the squares are needed.

1. **In words:** find the mean; for every value, take its distance from the mean and square it; add the squares; divide by the number of values (population) or by one less (sample).
2. **Formula:**

$$\sigma^2 = \frac{1}{N} \sum_{i=1}^N (x_i - \mu)^2 \qquad s^2 = \frac{1}{n-1} \sum_{i=1}^n (x_i - \bar{x})^2$$

σ^2 is the population variance, s^2 the sample variance.

3. **Example:** for 3, 2, 1, 5, 4 the mean is 3, and the distances are 0, -1 , -2 , 2, 1 (Figure 1). Their squares add up to $0 + 1 + 4 + 4 + 1 = 10$. As a population,

$$\sigma^2 = \frac{10}{5} = 2$$

As a sample,

$$s^2 = \frac{10}{5-1} = 2.5$$

Variance grows with spread but is not the spread itself: it is in squared units. For $-5, 0, 5$ the population variance is 16.7; for $-10, 0, 10$, which is twice as spread out, it is 66.7, four times as much.

4.1 Why we square the distances

Key point: The plain distances from the mean always add up to zero, so we need squares (or absolute values) to stop them cancelling.

In Figure 1 the distances are 0, -1, -2, 2, 1: the values below the mean give negative distances, those above give positive ones, and they add up to 0. That is not a coincidence.

Extra: Proof that the distances from the mean always add up to zero.

1. **In words:** the mean is the balance point of the data, so the negative and positive distances cancel exactly.
2. **Formula:** since $\sum_{i=1}^n x_i = n\bar{x}$ by the definition of the mean,

$$\sum_{i=1}^n (x_i - \bar{x}) = \sum_{i=1}^n x_i - n\bar{x} = n\bar{x} - n\bar{x} = 0$$

3. **Example:** for 3, 2, 1, 5, 4: $0 + (-1) + (-2) + 2 + 1 = 0$.

Spread should add up, never cancel. Squaring makes every distance positive. The absolute value does the same, and gives the mean absolute deviation of Section 5.

4.2 Variance and outliers

Key point: Squaring makes a far-away value count enormously, so variance is very sensitive to outliers.

A value far from the mean has a large distance, and its square is larger still. Add 50 to the five values above: the data becomes 3, 2, 1, 5, 4, 50, the mean moves to 10.83, and the population variance jumps from 2 to 308.5. One value multiplied the variance by about 150.

4.3 Why the sample variance divides by $n - 1$

Key point: Dividing by n makes a sample's variance too small on average; dividing by $n - 1$ makes it right on average.

We compute s^2 from a sample to estimate the population's σ^2 . If we divided by n , the estimate would be too small on average. Dividing by $n - 1$, a correction called **Bessel's correction**, fixes this.

A tiny example shows it exactly. Take the population 1, 2, 3, 4. Its mean is 2.5 and its variance is

$$\sigma^2 = \frac{(-1.5)^2 + (-0.5)^2 + 0.5^2 + 1.5^2}{4} = \frac{5}{4} = 1.25$$

Now draw every possible sample of two values, picking with replacement: (1, 1), (1, 2), ..., (4, 4), 16 samples in all. For a sample (a, b) the mean is $(a + b)/2$ and both values lie $|a - b|/2$ from it, so the squared distances add up to $(a - b)^2/2$. The table counts the 16 samples by their difference:

Difference $ a - b $	Samples	Squared distances, summed	Divide by $n = 2$	Divide by $n - 1 = 1$
0	4	0	0	0
1	6	0.5	0.25	0.5
2	4	2	1	2
3	2	4.5	2.25	4.5
Average over 16 samples			0.625	1.25

Dividing by n gives 0.625 on average, half the true 1.25. Dividing by $n - 1$ gives exactly 1.25.

Figure 2 repeats the experiment on real data. We treat the 714 known Titanic ages as a population (variance 211) and draw 20,000 random samples of each size from 2 to 20.

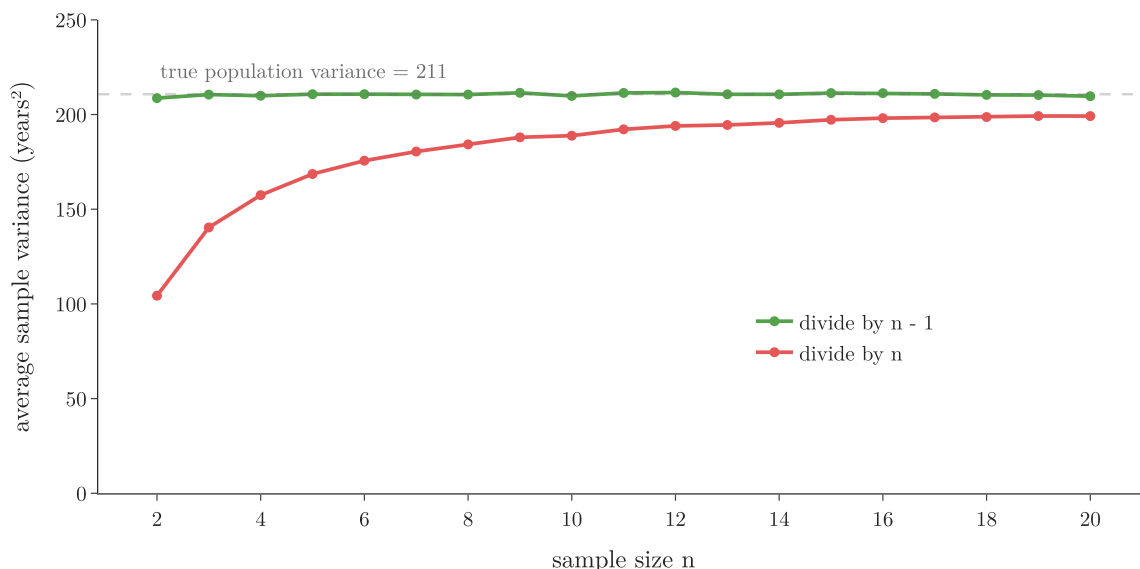


Figure 2: Average sample variance of Titanic ages, dividing by n or by $n - 1$

Dividing by $n - 1$ (green) lands on the true variance at every sample size. Dividing by n (red) is too small, badly so for small samples; the gap shrinks as n grows, because n and $n - 1$ become almost equal.

Extra: Why the sample is too narrow. The mean of a sample, $\bar{x}$, is the point closest to that sample's values: for any other point c , the squared distances are larger. The exact amount is an identity of algebra:

$$\sum_{i=1}^n (x_i - \mu)^2 = \sum_{i=1}^n (x_i - \bar{x})^2 + n(\bar{x} - \mu)^2$$

So measuring distances from $\bar{x}$ instead of the true μ always loses the amount $n(\bar{x} - \mu)^2$.

For 3, 2, 1, 5, 4 and $\mu = 2.5$: the left side is 11.25, and the right side is $10 + 5 \times 0.5^2 = 10 + 1.25 = 11.25$. Averaged over all samples, the lost amount equals exactly one σ^2 , which leaves $(n - 1)\sigma^2$: dividing by $n - 1$ gives back σ^2 . The step “equals one σ^2 ” needs the variance of a sample mean, which comes with the central limit theorem in a later maths Note.

Python: Population and sample variance.

```
import numpy as np
```

```
x = np.array([3, 2, 1, 5, 4])
```

```
np.var(x)          # 2.0: divides by n (ddof=0)
```

```
np.var(x, ddof=1)  # 2.5: divides by n - 1
```

ddof (“delta degrees of freedom”) is the number subtracted from n . NumPy uses ddof=0 by default; pandas’ `.var()` and `.std()` use ddof=1.

5. Mean absolute deviation

Key point: The mean absolute deviation averages the absolute distances from the mean; it is less sensitive to outliers than variance, but harder to work with mathematically.

The mean absolute deviation replaces the square by an absolute value (see the [PCA intuition Note](#), section 5.4).

1. **In words:** for every value, take its distance from the mean without the sign, then average these distances.
2. **Formula:**

$$\text{MAD} = \frac{1}{n} \sum_{i=1}^n |x_i - \bar{x}|$$

3. **Example:** for 3, 2, 1, 5, 4 the absolute distances are 0, 1, 2, 2, 1, so

$$\text{MAD} = \frac{0 + 1 + 2 + 2 + 1}{5} = \frac{6}{5} = 1.2$$

Its strength is that a far value counts in proportion to its distance, not its distance squared. So outliers inflate it less than they inflate variance.

Its weakness is mathematical. The absolute value has a sharp corner at zero, so formulas built on it cannot be differentiated there and are hard to manipulate. Variance has clean properties, such as the $n - 1$ correction above, so inferential statistics and most ML algorithms are built on variance. The mean absolute deviation is therefore rarely used.

Extra: The abbreviation MAD is also used for the median absolute deviation, a different, even more robust measure (see the [Pandas Profiling Note](#)). Always check which one is meant.

6. Standard deviation

Key point: The standard deviation is the square root of the variance; unlike the variance, it is in the same units as the data.

The standard deviation is the square root of the variance, $\sigma = \sqrt{\sigma^2}$ for a population and $s = \sqrt{s^2}$ for a sample. The [understanding your data Note](#) (section 7.1) works one through.

If we already have the variance, why keep the standard deviation? Because of units.

1. **In words:** take the square root of the variance, which brings the squared units back to the units of the data.
2. **Formula:**

$$\sigma = \sqrt{\frac{1}{N} \sum_{i=1}^N (x_i - \mu)^2}$$

3. **Example:** four people earn 16, 17, 13 and 14 LPA (lakh rupees per annum). The mean is 15 LPA, the distances are 1, 2, -2, -1 LPA, and

$$\sigma^2 = \frac{1^2 + 2^2 + (-2)^2 + (-1)^2}{4} = \frac{10}{4} = 2.5 \text{ LPA}^2$$

$$\sigma = \sqrt{2.5} \approx 1.58 \text{ LPA}$$

“2.5 LPA squared” means nothing to anyone: nobody earns a squared rupee. “1.58 LPA” does: a typical salary here is about 1.58 lakh away from the mean of 15 lakh. For 3, 2, 1, 5, 4, the standard deviation is $\sqrt{2} \approx 1.41$.

7. Coefficient of variation

Key point: The coefficient of variation divides the standard deviation by the mean, so we can compare the spread of columns measured in different units.

The standard deviation of salaries (in lakhs) and of experience (in years) cannot be compared: they are in different units, like apples and oranges. Yet we may want to know which of the two columns is more spread out relative to its own centre. The coefficient of variation (CV) answers this; the [Pandas Profiling Note](#) introduced it.

1. **In words:** divide the standard deviation by the mean, and multiply by 100 to get a percentage. The units cancel.
2. **Formula:**

$$\text{CV} = \frac{\sigma}{\mu} \times 100\%$$

3. **Example:** for the 714 known Titanic ages, the mean is 29.70 years and the standard deviation 14.53 years; for the 891 fares, the mean is 32.20 and the standard deviation 49.69. Then

$$\text{CV}_{\text{Age}} = \frac{14.53}{29.70} \times 100\% \approx 48.9\%$$

$$\text{CV}_{\text{Fare}} = \frac{49.69}{32.20} \times 100\% \approx 154.3\%$$

The fares vary about three times as much as the ages, relative to their means. Figure 3 shows why: dividing each column by its own mean puts both on one scale, where 1 is the mean. The ages stay within about 0 to 2.7 times their mean; the fares run from 0 to 16 times theirs.

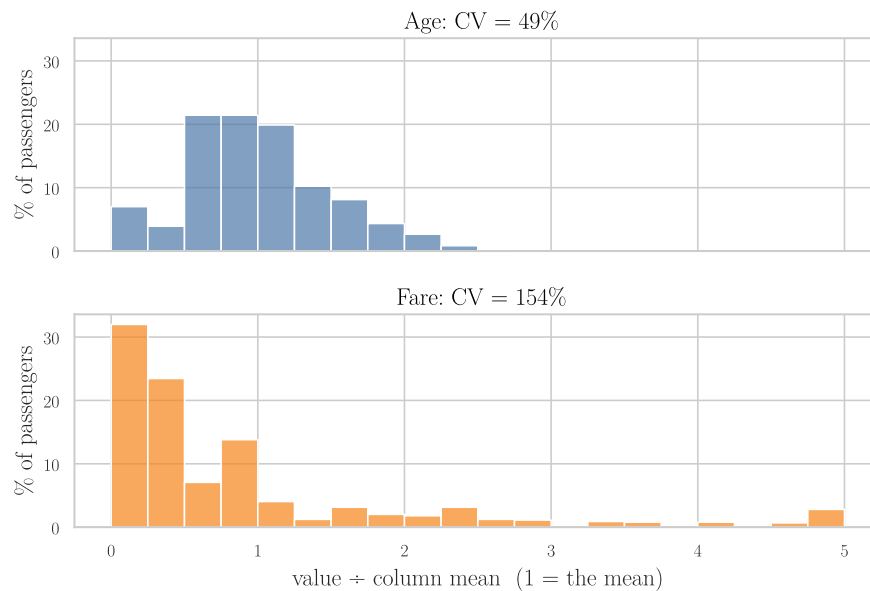


Figure 3: Titanic ages and fares, each divided by its own mean

The bigger the CV, the further the data spreads from its mean; the smaller, the more it is concentrated near the mean. The same idea compares two people's scores on different exams, or the spread of the same quantity in different currencies.

Extra: The CV only makes sense for data measured from a true zero and with a positive mean, such as fares, ages or weights. For temperature in Celsius, where 0 is arbitrary, or for a column with a mean near 0, the CV is meaningless or explodes.

Python: The coefficient of variation.

```
import pandas as pd
```

```
df = pd.read_csv("data/titanic_train.csv")
for col in ["Age", "Fare"]:
    cv = df[col].std() / df[col].mean() * 100
    print(col, round(cv, 1))    # Age 48.9, Fare 154.3
```

The Notebook for this Note ([notebook.ipynb](#)) computes every measure of this Note on the Titanic data and reruns the $n - 1$ experiment.

8. Summary

Measure	Formula	Example (3, 2, 1, 5, 4)	Units	Outliers
Range	$\max - \min$	4	data units	ruins it
Variance	$\sum (x_i - \mu)^2 / N$; sample divides by $n - 1$	2 (sample: 2.5)	squared	very sensitive
Mean absolute deviation	$\sum x_i - \bar{x} / n$	1.2	data units	less sensitive
Standard deviation	$\sqrt{\text{variance}}$	1.41	data units	sensitive
Coefficient of variation	$\sigma / \mu \times 100\%$	47%	none	sensitive

- Distances from the mean always add up to 0, so variance squares them.
- The sample variance divides by $n - 1$ (Bessel's correction); dividing by n is too small on average.
- The standard deviation is the variance brought back to the data's units.
- The CV compares the spread of columns in different units.

9. Key terms

Term	Meaning
Measure of dispersion	A number that describes how spread out a column is around its centre
Range	The largest value minus the smallest
Bessel's correction	Dividing by $n - 1$ instead of n , so the sample variance is right on average
ddof	NumPy and pandas argument: the number subtracted from n in the variance's denominator